

Emily Tan

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EDUCATION

University of California, Riverside

Bachelor of Science in Computer Science (Honors) GPA: 3.93/4.00

Riverside, CA

Expected 2028

EXPERIENCE

Lead Software Engineer | Cutie Hack Github

Apr. 2026 – Present

Association for Computing Machinery @UCR

Riverside, CA

- Built and led development of software for Cutie Hack, an ACM hackathon serving 300+ participants, in an Agile environment; conducted 10+ code reviews to maintain code quality
- Collaborated with software and UI/UX leads to design and implement user interfaces

Web Developer Intern | Rides | SHPE | AEP

Jan. 2026 – Aug. 2026

Association for Computing Machinery @UCR

Riverside, CA

- Collaborated in an Agile, GitHub issue-based workflow to deliver weekly frontend features from Figma designs
- Developed scalable UI components using Next.js, TypeScript, and Tailwind CSS for production deployment
- Implemented responsive design patterns to ensure consistent performance across devices and screen sizes

Fullstack Developer Intern | GitHub

Apr. 2026 – Jun. 2026

Association for Computing Machinery @UCR

Riverside, CA

- Collaborated in a multidisciplinary team to develop a smart surveillance system with real-time video detection
- Focused on full-stack development using Next.js, React, and TypeScript to build live monitoring dashboard
- Integrated computer vision streaming pipelines into the frontend using Flask video feeds and MQTT messaging

Firmware Intern

Oct. 2025 – Present

Formula SAE Highlander Racing @UCR

Riverside, CA

- Programmed STM32 microcontroller to transmit and process CAN bus messages for real-time vehicle systems
- Redesigned control logic using state machines and DMA to improve system speed and reduce CPU load

PROJECTS

UCR Registration Extension | GitHub | Webstore Link

Apr. 2026 – Present

- Built a Chrome extension that aggregates course data from multiple sources into a unified registration interface
- Integrated MyClassGrades, UCR course difficulty data, and RateMyProfessors into a structured data pipeline
- Implemented SPA-aware DOM injection with service workers and Chrome storage APIs for persistent performance
- Deployed the extension, helping 100+ students determine class difficulty during peak registration season

Pill-E – BioEngineering Makeathon | GitHub

Apr. 2026

- Built the full embedded system for a smart pill dispenser as the sole embedded engineer in a 3-person team
- Developed control logic for precise pill dispensing and real-time device state management
- Implemented device-to-mobile communication layer to interface with a companion app built by teammates
- Debugged and validated hardware performance through iterative testing and calibration of the dispensing system

DevTunes – Google DeepMind Hackathon | GitHub

Mar. 2026

- Worked in a 3-person team to build an AI system that analyzes source code to generate context-aware soundtracks
- Led project documentation, including system design write-ups, workflow diagrams, and technical specifications
- Contributed to designing modular workflows for code parsing, scene detection, and automated prompt generation

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, JavaScript

Web Development: React, Next.js, Node.js, HTML/CSS, Tailwind CSS

Embedded: STM32, CAN Bus, ADC, DMA, IoT, Arduino, Raspberry Pi

Data Science: pandas, NumPy, Matplotlib, scikit-learn

Developer Tools: Git, GitHub, VS Code, Continuous Integration, Scrum, Agile